

Mathematics Bootcamp — Day 1 Exercises

Mas-Colell, Whinston and Green, Chapter 1

Miguel Vázquez-Carrero

Preference and Choice: Sections 1.B and 1.C

Compilation Date: August 31, 2026

Notation

Throughout, X is a set of alternatives and $\succeq$ is a binary relation on X , read “at least as good as”. From it we derive

$$x \succ y \iff x \succeq y \text{ and not } y \succeq x, \quad x \sim y \iff x \succeq y \text{ and } y \succeq x.$$

The relation $\succeq$ is *rational* if it is *complete* (for all $x, y \in X$, $x \succeq y$ or $y \succeq x$) and *transitive* (for all $x, y, z \in X$, if $x \succeq y$ and $y \succeq z$ then $x \succeq z$). A function $u : X \rightarrow \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \iff u(x) \geq u(y) \quad \text{for all } x, y \in X.$$

A *choice structure* is a pair $(\mathbb{B}, C(\cdot))$: $\mathbb{B}$ is a family of nonempty subsets of X , the observed menus, and C assigns to each $B \in \mathbb{B}$ a nonempty set $C(B) \subseteq B$. MWG writes $\mathcal{B}$ for the menu domain; we keep the $\mathbb{B}$ of the slides.

Definition 1.C.1 (Weak Axiom). If for some $B \in \mathbb{B}$ with $x, y \in B$ we have $x \in C(B)$, then for any $B' \in \mathbb{B}$ with $x, y \in B'$ and $y \in C(B')$, we must also have $x \in C(B')$.

Definition 1.C.2 (Revealed preference). $x \succeq^* y \iff$ there is some $B \in \mathbb{B}$ such that $x, y \in B$ and $x \in C(B)$.

Exercises 1.B.1 and 1.B.2 ask for the proof of the following, which MWG states and does not prove.

Proposition 1.B.1. If $\succeq$ is rational then:

- (i) $\succ$ is both *irreflexive* ($x \succ x$ never holds) and *transitive* (if $x \succ y$ and $y \succ z$, then $x \succ z$).
- (ii) $\sim$ is *reflexive* ($x \sim x$ for all x), *transitive* (if $x \sim y$ and $y \sim z$, then $x \sim z$), and *symmetric* (if $x \sim y$, then $y \sim x$).
- (iii) if $x \succ y \succeq z$, then $x \succ z$.

How to write these up

Write every proof the way a program is written: the skeleton first, then the body. Recall the skeleton from Day 1,

Let x be arbitrary.

Suppose $P(x)$ is true.

[Proof of $Q(x)$ goes here.]

Thus, if $P(x)$ then $Q(x)$.

Thus, for all x , if $P(x)$ then $Q(x)$.

and note that the indentation is not decoration. A line is available exactly to the lines indented under it, and a block is closed by the line that returns to the outer margin. Write the outer lines first: they are forced by the shape of the statement alone, before you have any idea why it is true.

Before any of that, state what the proof owes. Unfold every definition in the conclusion until what is left is a list of claims with no defined terms in them. That list is the **Need to Show**. A proof is finished when the list is exhausted, and not before.

Preferences (Section 1.B)

Exercise 1.B.1 (grade B)

Property (iii) of Proposition 1.B.1

Prove property (iii) of Proposition 1.B.1.

Exercise 1.B.2 (grade A)

Properties (i) and (ii) of Proposition 1.B.1

Prove properties (i) and (ii) of Proposition 1.B.1.

Exercise 1.B.3 (grade B)

A strictly increasing transformation of a utility function

Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function and $u : X \rightarrow \mathbb{R}$ is a utility function representing preference relation $\succeq$, then the function $v : X \rightarrow \mathbb{R}$ defined by $v(x) = f(u(x))$ is also a utility function representing preference relation $\succeq$.

Exercise 1.B.4 (grade A)

A sufficient condition for representation

Consider a rational preference relation $\succeq$. Show that if $u(x) = u(y)$ implies $x \sim y$ and if $u(x) > u(y)$ implies $x \succ y$, then $u(\cdot)$ is a utility function representing $\succeq$.

Exercise 1.B.5 (grade B)

Representation on a finite set

Show that if X is finite and $\succeq$ is a rational preference relation on X , then there is a utility function $u : X \rightarrow \mathbb{R}$ that represents $\succeq$. [Hint: Consider first the case in which the individual's ranking between any two elements of X is strict (i.e., there is never any indifference), and construct a utility function representing these preferences; then extend your argument to the general case.]

Choice Rules (Section 1.C)

Exercise 1.C.1 (grade B)

What the weak axiom leaves open

Consider the choice structure $(\mathbb{B}, C(\cdot))$ with $\mathbb{B} = \{\{x, y\}, \{x, y, z\}\}$ and $C(\{x, y\}) = \{x\}$. Show that if $(\mathbb{B}, C(\cdot))$ satisfies the weak axiom, then we must have $C(\{x, y, z\}) = \{x\}$, $= \{z\}$, or $= \{x, z\}$.

Exercise 1.C.2 (grade B)

A symmetric restatement of the weak axiom

Show that the weak axiom (Definition 1.C.1) is equivalent to the following property holding:

Suppose that $B, B' \in \mathbb{B}$, that $x, y \in B$, and that $x, y \in B'$. Then if $x \in C(B)$ and $y \in C(B')$, we must have $\{x, y\} \subseteq C(B)$ and $\{x, y\} \subseteq C(B')$.

Exercise 1.C.3 (grade C)

Two revealed strict preferences

Suppose that choice structure $(\mathbb{B}, C(\cdot))$ satisfies the weak axiom. Consider the following two possible revealed preferred relations, $\succ^*$ and $\succ^{**}$:

$$\begin{aligned} x \succ^* y &\iff \text{there is some } B \in \mathbb{B} \text{ such that } x, y \in B, x \in C(B), \text{ and } y \notin C(B) \\ x \succ^{**} y &\iff x \succeq^* y \text{ but not } y \succeq^* x \end{aligned}$$

where $\succeq^*$ is the revealed at-least-as-good-as relation defined in Definition 1.C.2.

- (a) Show that $\succ^*$ and $\succ^{**}$ give the same relation over X ; that is, for any $x, y \in X$, $x \succ^* y \iff x \succ^{**} y$. Is this still true if $(\mathbb{B}, C(\cdot))$ does not satisfy the weak axiom?
- (b) Must $\succ^*$ be transitive?
- (c) Show that if $\mathbb{B}$ includes all three-element subsets of X , then $\succ^*$ is transitive.